1 Data Collection & Import

- **Identify the data source:** Databases, CSV/Excel files, APIs, or manual entry.
- **Import into Excel:** Use *Get Data* (Power Query) to pull data from CSV, SQL Server, web, or other files.
- **Check file formats:** Ensure consistent delimiters, encoding (UTF-8), and column headers.

2 Data Cleaning & Preparation

Cleaning is the most critical step :

- **Remove duplicates:** Data → Remove Duplicates or use UNIQUE() formula.
- **Handle missing values:** Replace blanks using IFERROR, IFNA, or Power Query transformations.
- **Standardize data:** Trim spaces (TRIM()), correct text case (PROPER()), fix date/time formats.
- **Data types:** Make sure numbers, text, and dates are properly formatted.
- **Split/merge columns:** Text to Columns, CONCAT(), or Power Query *Split Column*.

3 Data Transformation

- **Create calculated columns:** Use formulas (IF, VLOOKUP/XLOOKUP, INDEX-MATCH, TEXT, DATEDIF).
- **Apply conditional logic:** IF, IFS, SWITCH for categorizing data.
- **Aggregation:** Use SUMIFS, COUNTIFS, AVERAGEIFS for grouped metrics.
- **Power Query transformations:** Unpivot columns, merge/join multiple tables, append datasets

4 Exploratory Data Analysis (EDA)

Understand the story your data is telling :

- **Descriptive statistics:** AVERAGE, MEDIAN, STDEV.P, MAX, MIN.
- **Sort & filter:** Quick checks for trends/outliers.
- **Conditional formatting:** Highlight top/bottom performers, outliers, or thresholds.
- **Quick charts:** Column, bar, line, scatter, histogram for visual inspection.

5 Data Modelling & Pivot Tables

- **Create relationships:** Use *Data Model* to relate multiple tables (like a database).
- **Pivot Tables:** Drag and drop fields to summarize data dynamically.
- **Pivot Charts:** Visualize metrics with interactive filters (Slicers, Timelines).
- **Calculated Fields/Items:** Add KPIs like profit margins or YoY growth.

6 Advanced Analysis

- **Scenario analysis:** What-If Analysis, Goal Seek, Solver for optimization.
- **Trend analysis:** Moving averages, forecasting using FORECAST.ETS.
- **Correlation & Regression:** Use *Data Analysis ToolPak* for deeper statistics.

7 Data Visualization & Dashboarding

- **Clean design:** Use consistent colors, data labels, and legends.
- **Interactive elements:** Slicers, drop-downs, buttons for user-driven insights.
- **Dynamic charts:** Combine Pivot Charts and named ranges for automation.

8 Reporting & Sharing

- **Insights summary:** Highlight key metrics, trends, and recommendations.
- **Export options:** Save as Excel, PDF, or share via OneDrive/SharePoint.
- **Automation:** Use Macros/VBA or Power Query refresh for recurring reports.